



Azure DocumentDB

Develop AI-powered, MongoDB-compatible apps and standardize your NoSQL data with an open-source, multicloud database



What Is DocumentDB?

The **open-source** MongoDB-compatible DB, built for flexibility, scale, and AI.

Now part of the **Linux Foundation**

Broad Industry Support: Azure, AWS, GCP, Snowflake, Cockroach Labs, Yugabyte, Supabase, Ubicloud, and more..

Reference Implementation of MongoDB API: Like Postgres for SQL. DocumentDB brings an open standard to Document workloads.

Multi-Cloud by Design: Use it anywhere: Azure, AWS, GCP, on-prem. No vendor lock-in.

Runs the Azure DocumentDB Offering: Powering our managed service under the hood.

"AWS is excited to contribute to the open source DocumentDB project, now stewarded by the Linux Foundation." – **Adam Abrevaya, Director, AWS DocumentDB**

"It is great to see the DocumentDB project joining the Linux Foundation which assures customers and the community have an openly-governed, open-source option available to them." – **Sailesh Krishnamurthy, VP, Google Cloud**

The Data Challenge of Modern Apps

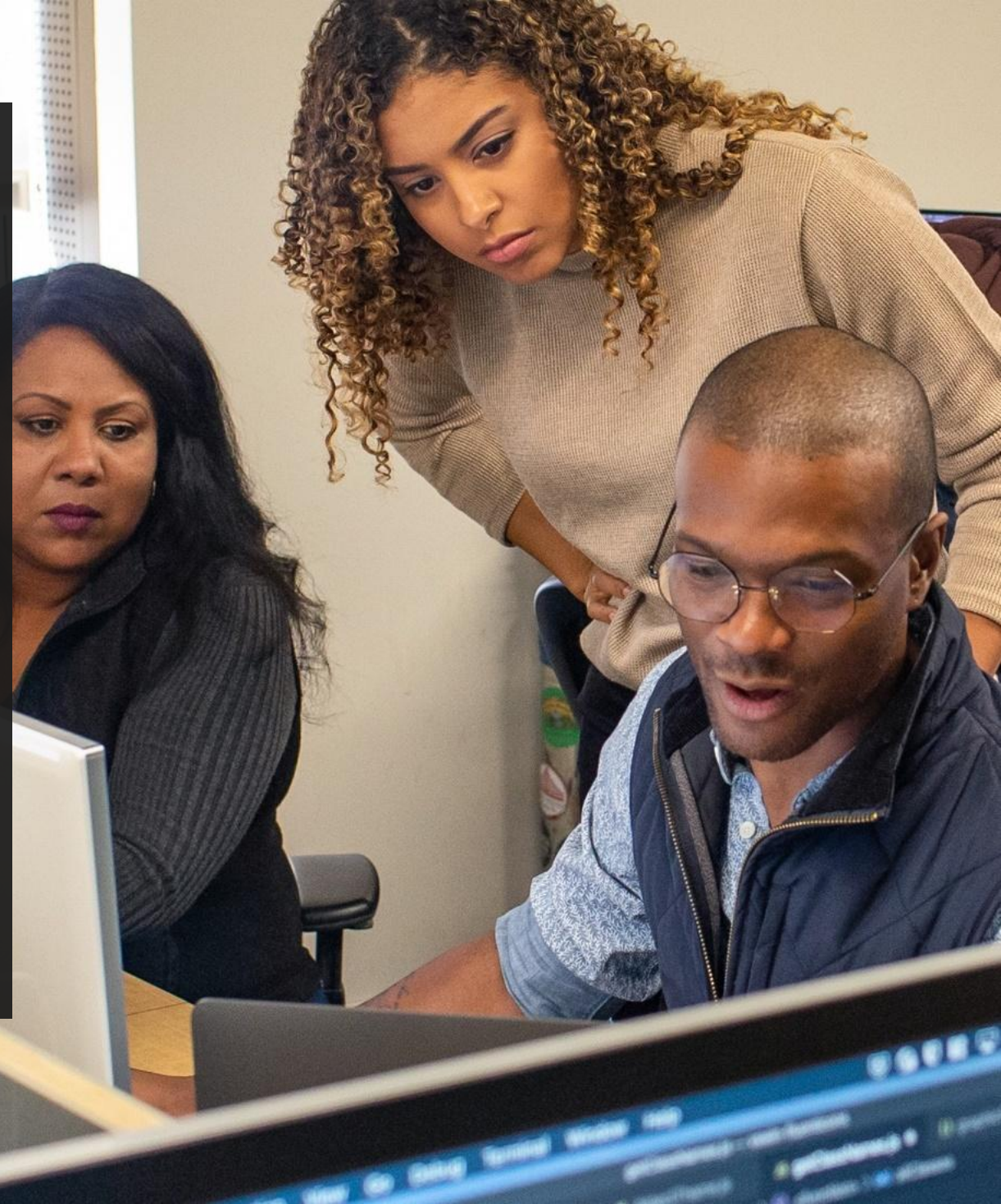
- Customers are looking to **optimize their scale-up architecture** for richer queries and transactions while using a **flexible schema**
- Modern apps demand **flexibility, intelligence, and freedom across clouds** – but data systems haven't kept up
- With budgets tightening, **enterprise costs are facing unprecedented scrutiny.**
- Organizations running MongoDB-compatible workloads want to **standardize their NoSQL databases** with open source while ensuring security, performance, and compliance across hybrid and multi-cloud environments



Many app developers today use MongoDB or DocumentDB

- Allows to specify documents in JSON format
- JSON fits naturally into many web developer's code bases with the popularity of JavaScript
- Part of modern technology stacks such as MEAN & MERN widely used in web apps
- Popular skillset among developers

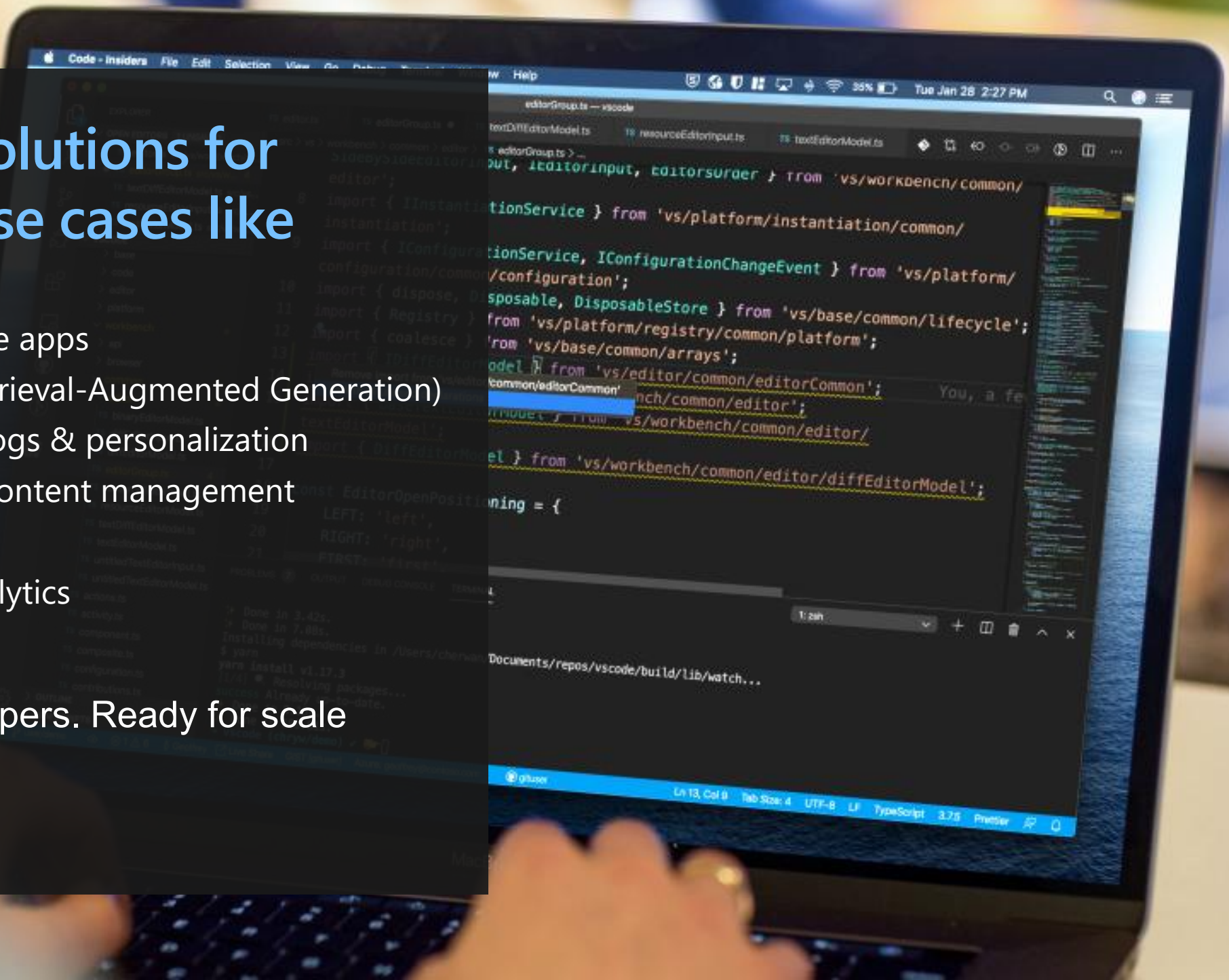
Large community ecosystem means faster time to value!



To build solutions for modern use cases like

- Web & Mobile apps
- AI + RAG (Retrieval-Augmented Generation)
- Product catalogs & personalization
- Inventory & content management
- IoT platforms
- Real-time analytics

Built for developers. Ready for scale



Non-relational Databases on Azure



Best resiliency, scalability, and performance for agentic apps

Azure Cosmos DB
(NoSQL API)

Optimized for Scale Out

Industry leading high availability (5x 9's)

Performance at any scale

Fastest Autoscale + Serverless

Why do customers care

95%+ of all customers have non-relational Databases, and are growing their usage to keep the pace of AI Innovation



Build MongoDB-compatible apps
Open Source, Hybrid/Multi-Cloud

Azure Document DB

Optimized for Scale Up

Familiar vCore model + MongoDB API

Open Source, Hybrid/Multi-Cloud

Rich query and transactions

Azure DocumentDB



Build innovative apps with open-source, **MongoDB-compatible** document database in any environment, deploy in Azure for the best enterprise experience with hybrid and multicloud capabilities



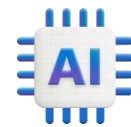
Open source and multicloud

- Based on **DocumentDB**, a **MongoDB-compatible open-source project**
- Backed by the Linux Foundation
- Develop anywhere, deploy at scale in Azure
- Speed up innovation using you existing skills, familiar tools, and drivers



Enterprise Ready

- Proven technology **(formerly vCore-based Azure Cosmos DB for MongoDB)**
- Authenticate with Microsoft Entra ID
- Protect data with Customer-Managed Keys
- Enterprise-grade reliability with up to 99.995% availability SLA across the full service stack



AI-driven

- Built-in, no-cost vector search powered by scalable DiskANN
- Combine vector and full-text with hybrid search
- Build AI apps and agents easily with seamless integration with Azure ecosystem



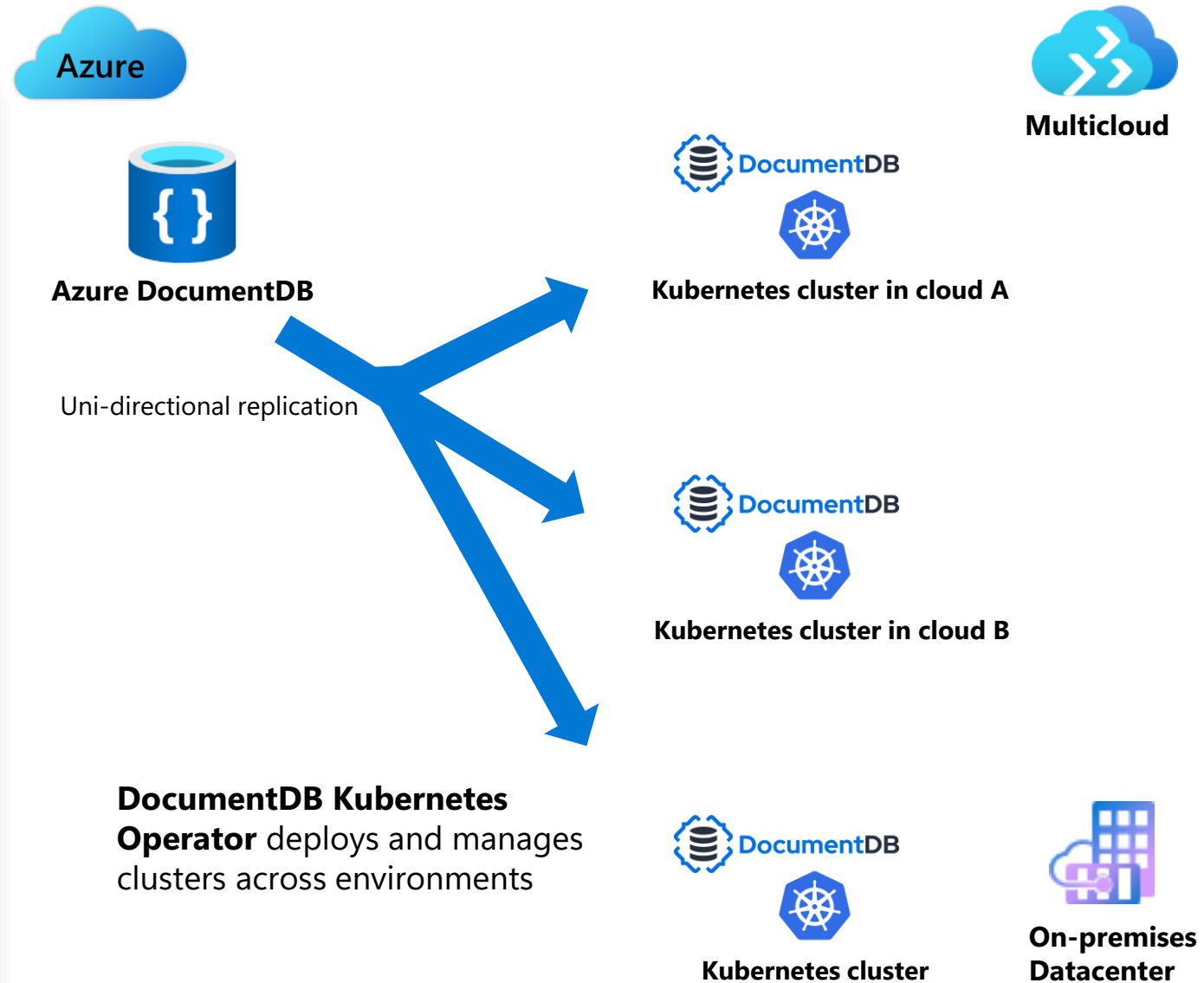
Cost effective

- Reduce total cost of ownership by scaling compute and storage independently, 35 days of free backups, and 24/7 support included in price
- Avoid overprovisioning with automatic and instant scaling
- Get started for free with Open Source or free tier option

Hybrid and multicloud freedom with Azure DocumentDB

Key features

- **One open engine, many environments** – Azure DocumentDB and the open-source DocumentDB engine share the same foundation
- **Kubernetes-native deployment anywhere** – Use the DocumentDB Kubernetes Operator to deploy, run and manage clusters across AWS, GCP, and on-premises
- **Continuous sync across clouds** – Replicate between managed Azure DocumentDB and self-hosted clusters for hybrid/multicloud, or DR topologies
- **Consistent developer experience** – Same MongoDB-compatible API, drivers, and tools
- **Hybrid continuity and resilience** – Build globally distributed or cross-cloud architectures for compliance, sovereignty, or failover



Home > CosmosDB-MongoDB-azure-documentdb-226ceb07a25d4b46a703937f016f78 | Overview > alaye-test-rg

azure-documentdb

Azure DocumentDB (with MongoDB compatibility)

☆

...

How do I troubleshoot performance issues with this resource?

List connection strings for this Azure DocumentDB.

+1

Search

Refresh

Reset password

Update

Delete

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Quick start

Resource visualizer

Settings

Monitoring

Automation

Help

Essentials

Status

Ready

Resource group (move)

alaye-test-rg

Subscription (move)

Project Everest Test Subscription

Subscription ID

81901d5e-31aa-46c5-b61a-537dbd5df1e7

Location (move)

East US 2

MongoDB version

8.0

Admin username

alaye

Cluster tier

M30 tier, 2 vCores, 8 GiB RAM

Disk size

128 GiB

Shard count

1

High availability

Yes

Connectivity method

Public access

Authentication

Native DocumentDB only

Storage encryption

Service-managed key

Read region

Not enabled

View Cost

JSON View

Add or remove favorites by pressing Ctrl+Shift+F

DocumentDB vs MongoDB Atlas

Pay As You Go
Public Pricing

Additional RI discounts
available for 1yr (40% off)
and 3yr (60% off)

Cluster Tier	Storage	RAM	vCPUs	Azure DocumentDB	MongoDB Atlas	% Diff
M10	32GB	2GB	1	\$0.03	\$0.10	-70%
M20	64GB	4GB	2	\$0.12	\$0.23	-48%
M30	128GB	8GB	2	\$0.32	\$0.62	-48%
M40	256GB	16GB	4	\$0.65	\$1.21	-46%
M50	512GB	32GB	8	\$1.31	\$2.38	-45%
M60	1TB	64GB	16	\$2.62	\$4.49	-42%
M80	1TB	128GB	32	\$4.91	\$8.20	-40%
M200	2TB	256GB	64	\$8.99	\$15.05	-40%

East US Azure DocumentDB vs MongoDB Atlas price comparison by cluster tier

Storage Savings – Decoupled from Compute cost

Separating the cost of Storage and Compute

- Compute and Storage can be scaled independently.
- Storage heavy workloads can have small nodes with large disks.
- Compute heavy workloads can have large nodes with small disks.
- Compute and Storage heavy workloads can have large VMs and disks to optimize both.

	Storage in GB	# nodes needed	Compute Cost	Storage Cost	Total Cost
Max Storage per node – competitors	4,000	100	\$129,125	\$47,100	\$176,225
Max Storage per node – vCore today	32,000	13	\$16,141	\$5,888	\$22,028, \$44,056 (x2)
Max Storage per node – vCore in 4-6 months	64,000	6	\$8,070	\$2,944	\$11,014 \$22,028 (x2)

Azure DocumentDB supports upto 32TB disks today, compared to MongoDB Atlas which only supports upto 4TB disks.

Storage heavy workloads, in the order of 10s or 100s of TBs will see a massive cost increase on Atlas, due to the minimum number of nodes needed to sustain the storage requirements.

Larger disks on Azure DocumentDB lead to significantly lower costs when running at scale.

Lower your TCO with Azure DocumentDB

- Lower total cost of ownership (TCO)
 - **45% lower cost** for comparable cluster tier compute/storage (public pricing)
 - **Additional 50% off** for dev/test
 - **1yr (40% off) and 3yr (60% off) RI discounts** available.
- Pay only for what you use by scaling compute and storage independently
- Better throughput and latency performance for comparable cluster tier
- Up to 32TB per shard
- Free point-in-time backup and restore (35d)
- No additional support contract/licenses

Save 50% switching from MongoDB Atlas to Azure DocumentDB



SSD v2 – Higher perf at the same cost



Cheaper. Better. Faster.

Premium **SSD v2** on Azure DocumentDB
→ **UP TO 80K IOPS!**

THE OLD MATH:

Month 1: 7K IOPS \$280/mo

Month 2: 10K IOPS \$1,110/mo

Month 3: 19K IOPS \$4,500/mo

Overprovisioned Storage

THE NEW MATH:

→ Month 1: 7K IOPS \$280/mo
80K IOPS

→ Month 2: 10K IOPS Same \$280/mo

→ Month 3: 18K+ IOPS Still \$280/mo

→ **4X Headroom!**

\$4,500 → \$280

12x Performance!

Decoupled Storage & Performance.

Previously, scaling from 7K IOPS to 20K IOPS meant increasing the size of the disk, even when additional storage was not needed.

With SSD v2:

1. IOPS is configured to the max at 80K
2. Bandwidth is configured to the max at 1200 MBps
3. IOPS and bandwidth are not charged

12x performance when moving to SSD v2

Same price as SSD v1

1 million writes per second – single node

Azure DocumentDB



\$5.20/hour

M80 Compute (\$4.41) + 2TB SSD v2 (\$0.77)



1M Writes/sec

(sustained)



192K Trans / \$

MongoDB Atlas on AWS



\$8.44/hour

M80 Compute + 2TB Disk

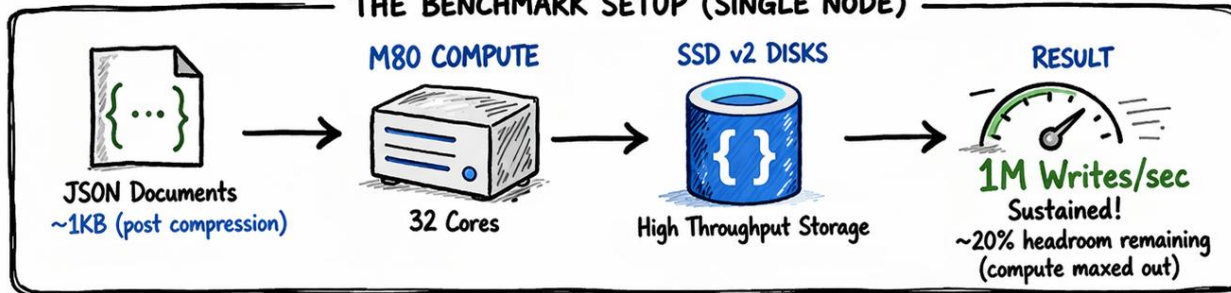


800K Writes/sec



95K Trans / \$

THE BENCHMARK SETUP (SINGLE NODE)



On Azure DocumentDB, we are able to achieve 1M writes per second on a single node.

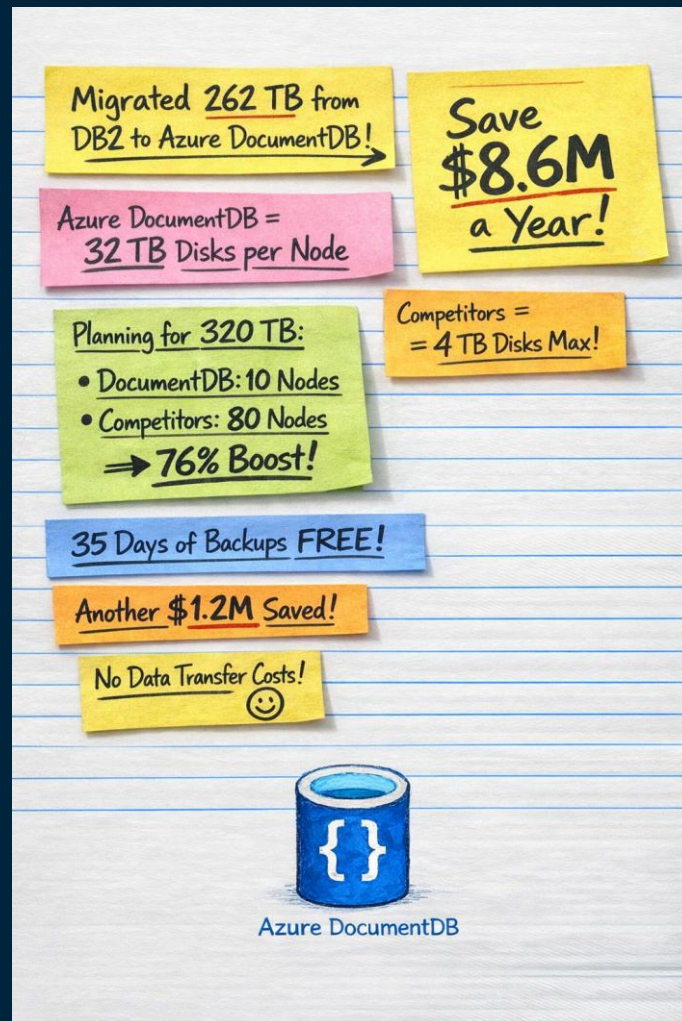
No sharding. No modeling complexities.

Compute and Storage costs are lower on Azure DocumentDB w.r.t. MongoDB Atlas

Azure DocumentDB is also able to achieve 20% more write throughput.

Transactions per dollar are 2x on Azure DocumentDB than they are on MongoDB Atlas.

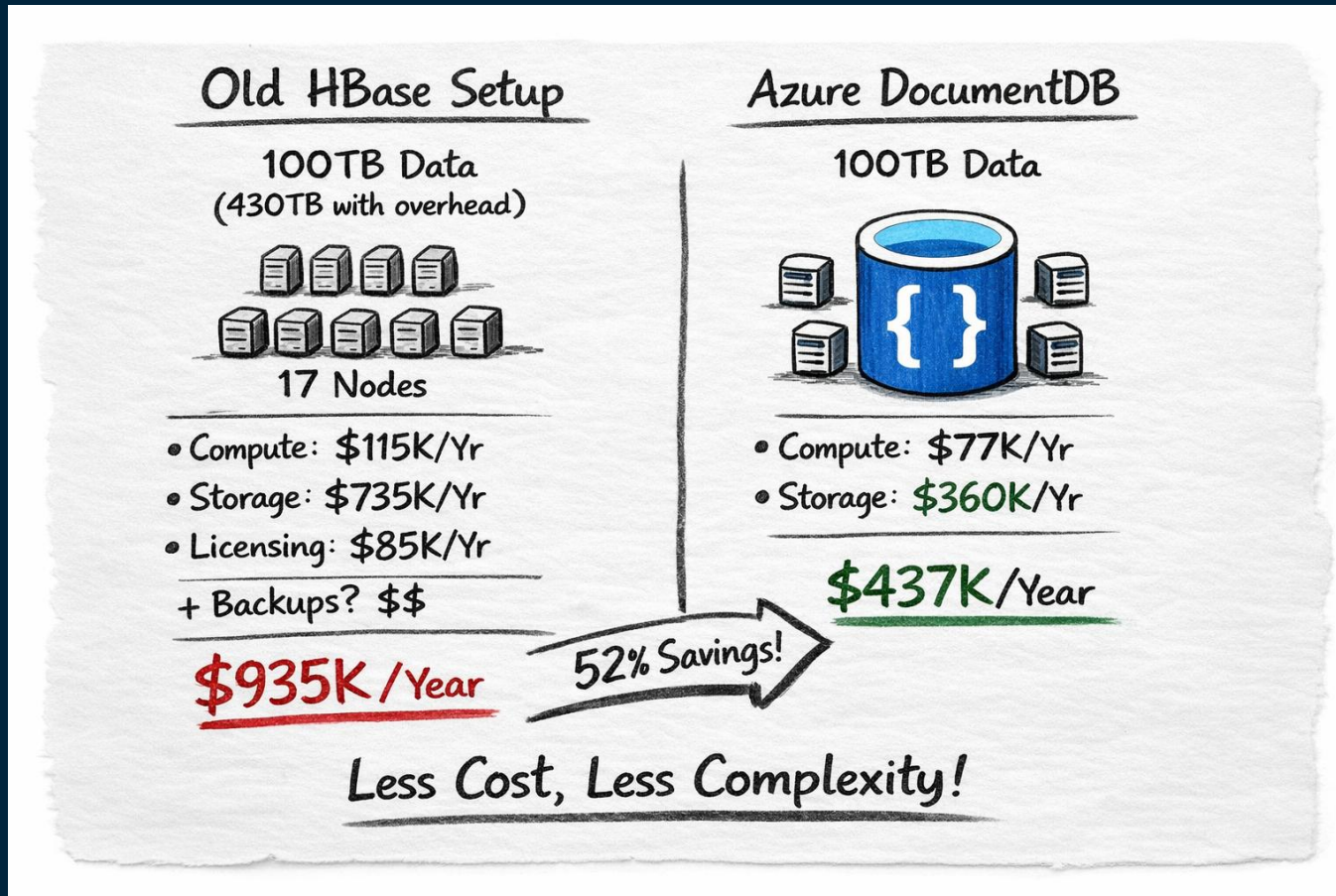
DB2 to Azure DocumentDB TCO Reduction



HealthCare Customer saved nearly \$9M a year when moving to Azure DocumentDB from DB2

1. No cost of backups
2. No networking charges for data ingress/egress
3. Large 32TB disks facilitated fewer node counts
4. Led to an 87% reduction in costs w.r.t. MongoDB Atlas

HBase - Azure DocumentDB Savings



Against MongoDB Atlas – it's clear the savings are significant on Azure DocumentDB.

Recently, we are seeing more Hbase migrations also moving to Azure DocumentDB.

Azure DocumentDB comes in at least 50% cheaper than Hbase – including Cloudera.

No licensing. No backup costs. No networking costs.

These add up particularly when running at scale.

Scaling with Azure DocumentDB



Vertical Scaling

Vertical scaling involves increasing resources like vCores and RAM of existing nodes or shards easily.



Horizontal Scaling

Horizontal scaling distributes data across multiple nodes, enabling support for larger datasets and high throughput.



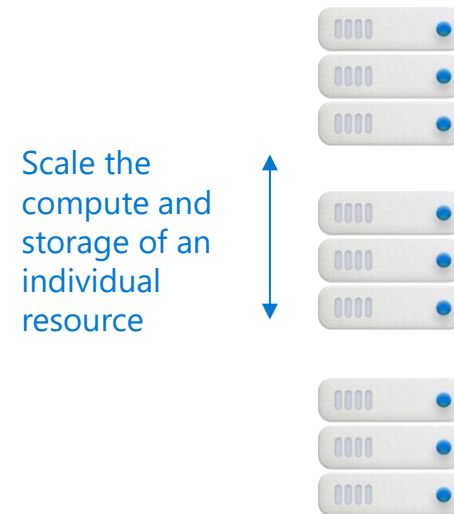
Compute storage Isolation

Storage scaling allows for independent disk size increases without altering the compute tier, providing flexibility.

Vertical (Scale-Up)

- Azure DocumentDB supports scaling compute (vCores/RAM) and storage/IOPS (disk size) **independently**
- You can **increase or decrease compute and increase storage without service downtime** or application changes
- Larger disks (**up to 32TB per shard**) allow fewer shards for large-storage workloads
- The architecture supports “smaller compute, larger disk” depending on workload

Vertical scaling (scale-up)



Scale out by provisioning more clusters



The **only** MongoDB with Instant AutoScale

Why Instant Autoscale?:

- Instant Scale: ensuring peak demand is always met instantly.
- Budget-friendly: Pay only for what you use.
- No Overprovisioning: Easy win for any customer that's overprovisioned

Autoscale cluster tier: can be chosen at any point, with zero downtime

Market leading scale-up/down: unlike MongoDB Atlas that takes hours to scale up and days to scale down.

Saves customer money: Threshold @35%, Min. Price for Normal Usage, Max. Price for surges. Predictable pricing with flexibility.

Encourages customers to choose the RIGHT size:

Customers will choose the right tier for their workload, avoid over/under provisioning, and reduce escalations.

Horizontal (Scale-Out)

- When scaling vertically is no longer sufficient, you can scale horizontally by increasing physical shard count
- **Azure DocumentDB manages logical -> physical shard mapping**
- Logical shards are mapped into physical shards and requests are routed automatically.
- **Rebalancing is handled behind the scenes**
- Adding or removing shards **does not require application changes** (same connection string) and does not change the existing mapping of logical to physical shards until necessary

Scale the
compute and
storage of an
individual
resource



Horizontal scaling

Scale out by adding more shards



Azure DocumentDB – Use cases

An open-source, MongoDB-compatible document database for hybrid and multicloud

MongoDB Migrations

App Modernizations

Gen AI New Apps



Largest MongoDB workload on Azure



Several lines of business with each very small

Migrations, modernizations, new apps



Few lines of business with each massive in scale

Modernizations over migrations and new apps



Gen AI focused new apps

Healthcare and Financial Services

Introducing

Premium SSD V2

At the same cost as Premium SSD

Azure DocumentDB gets you the Maximum IOPS and Throughput for every disk selected, at no additional fee.

Up to 80,000 IOPS(4x compared to SSDv1) and 1,200 MB/s

High Availability for Azure DocumentDB

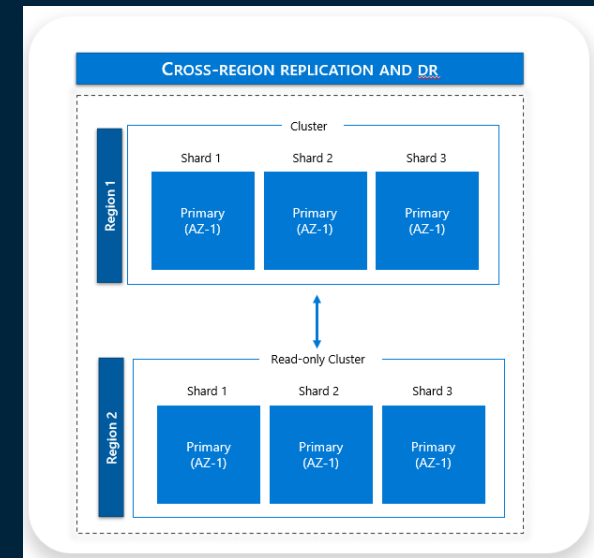
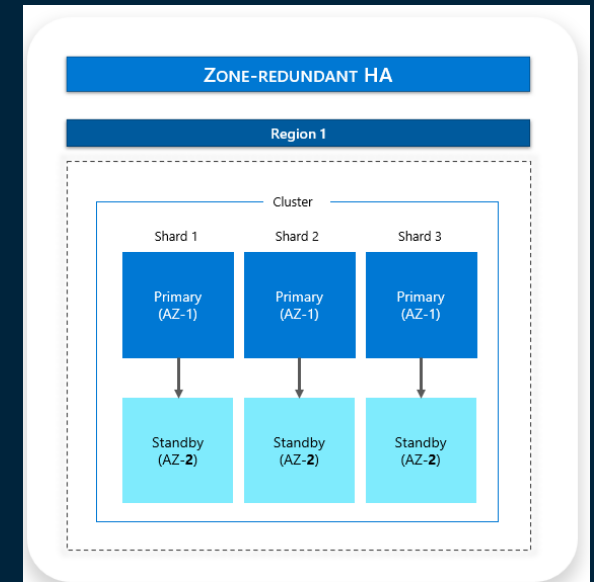
Azure DocumentDB has two primary HA configurations:

In-region high availability:

- When HA is enabled, each shard has a primary-standby physical shard pair within the same Azure region with synchronous replications (thus zero loss) and automatic failover.
- With in-region high availability enabled, **you can achieve a 99.99% monthly availability SLA for single-region clusters**

Cross-region replication with disaster recovery and read-scale:

- Azure DocumentDB supports replica clusters in another Azure region (or same region but different zone) for disaster recovery (DR) and read scalability. The replication is typically asynchronous (for cross-region) but can be synchronous.
- Best-practice documentation recommends **combining in-region HA with cross-region replication to achieve a 99.995% HA SLA**



Layers of security with Azure



Authentication and access management

Native user role password authentication



Data protection

Transparent data encryption at rest with Microsoft-managed keys



Network security

Private access via proven Private Link technology

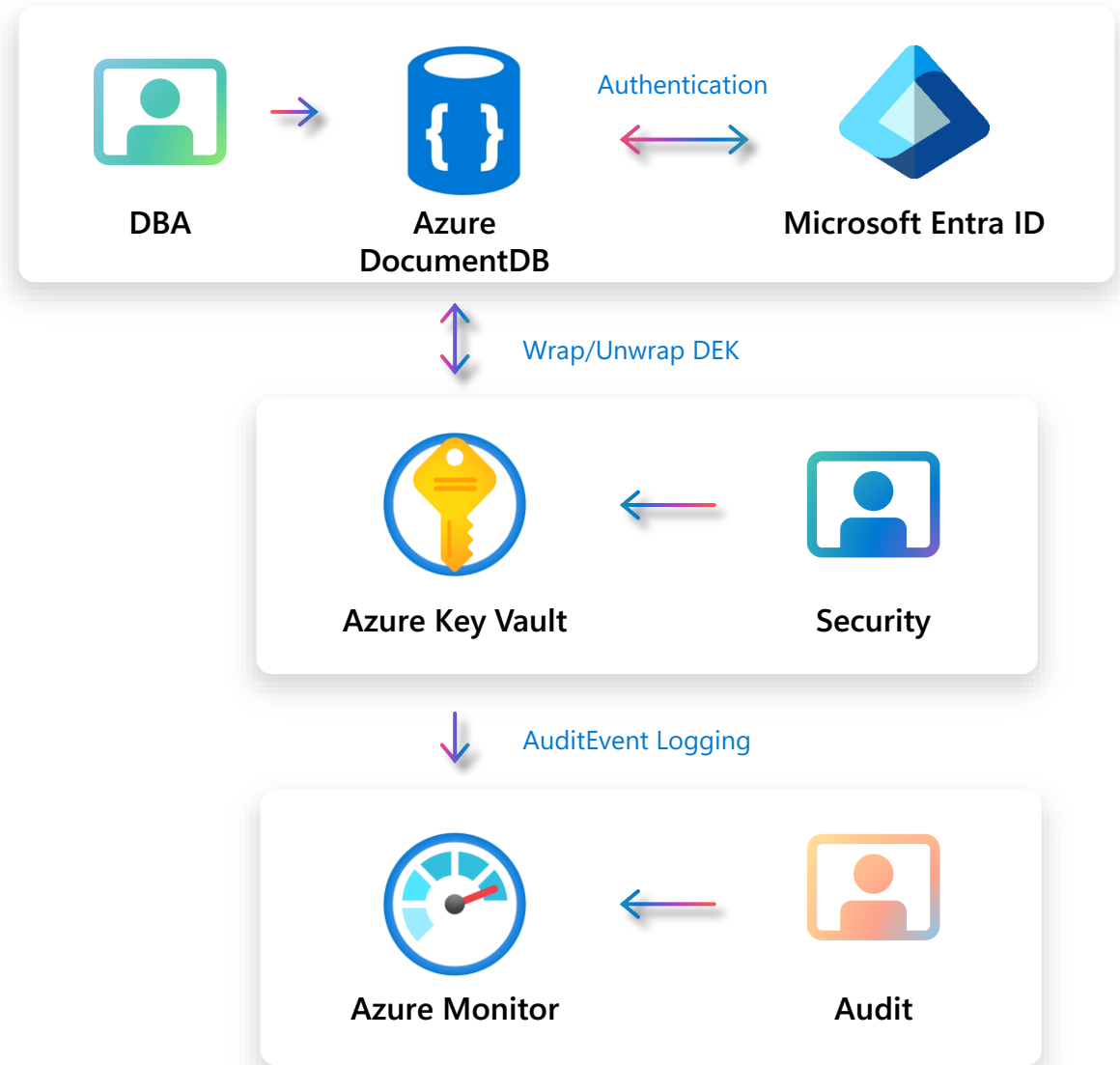
Public endpoints with Firewall Rules

Customers Managed Keys

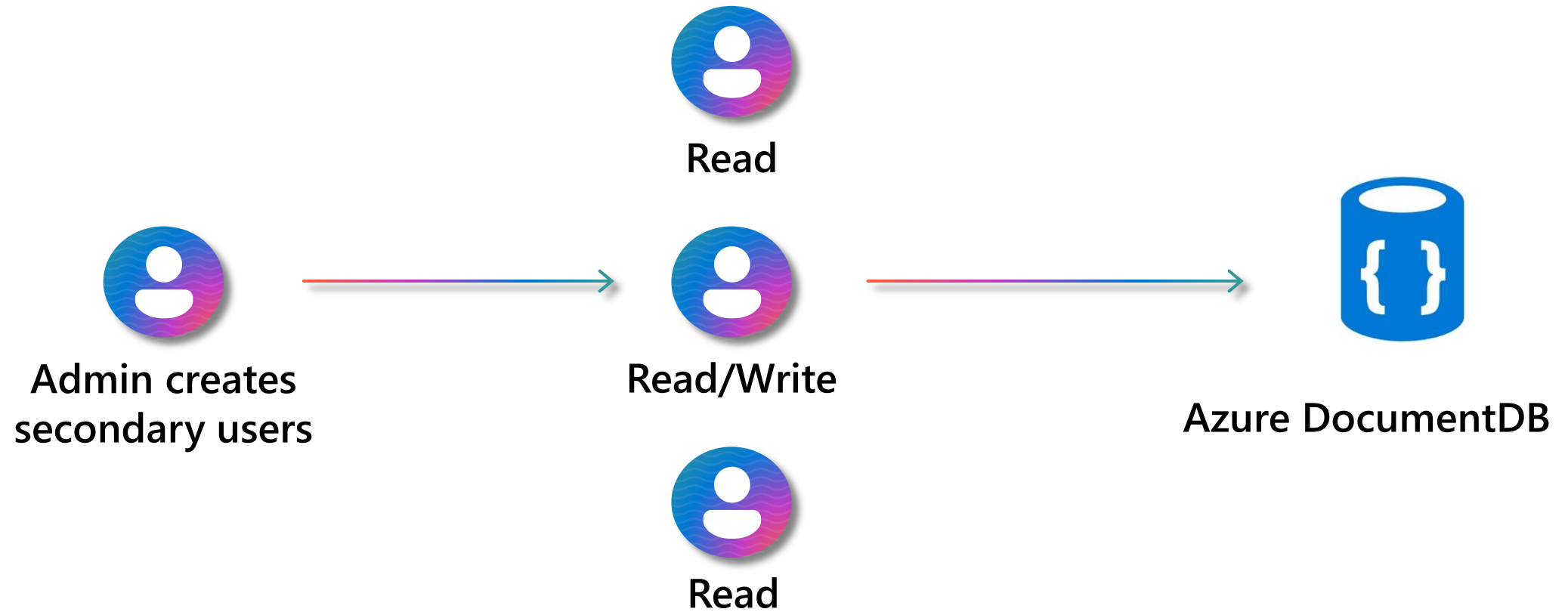
A Customer Managed Key (CMK) provides an additional encryption layer for data stored in the database

Benefits:

- You fully control data access. You can remove a key to make data inaccessible
- You fully control a key's life cycle, including rotation of the key, to align with corporate policies
- You can centrally manage and organize all your encryption keys in your own instances of Azure Key Vault
- Data encryption based on CMK doesn't have negative impact on the performance of your workload
- You can implement separation of duties between security officers, database administrators, and system



Role-Based Access Control for Secondary Users

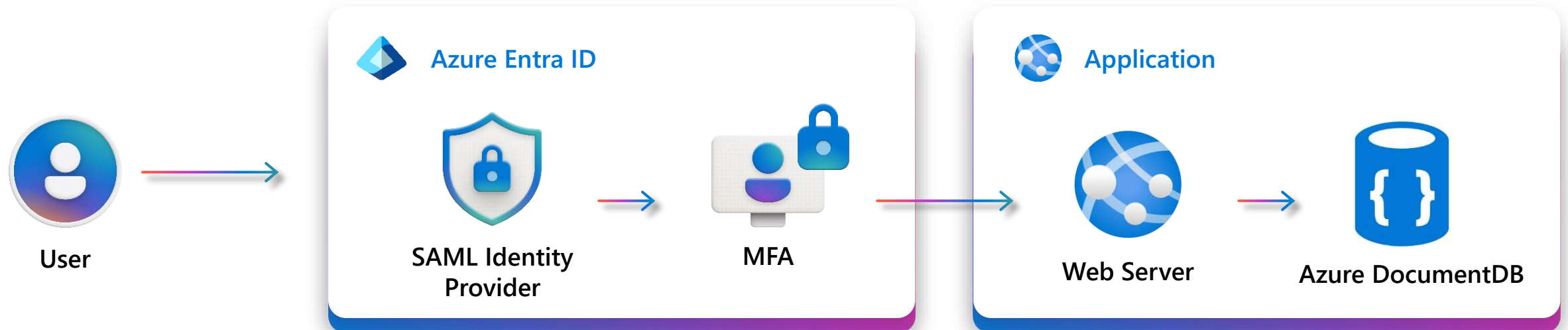


Entra ID authentication

Authenticate to Azure DocumentDB clusters using Entra ID accounts, providing a unified Azure identity management experience

Benefits:

- Authentication of users across Azure services in a uniform way
- Management of password policies and password rotation in a single place
- Multiple forms of authentication supported by Entra ID
- Support for token-based authentication for applications connecting to Azure DocumentDB



Monitoring and Insights in Azure DocumentDB

Enable full-stack visibility, alerting, and diagnostic for MongoDB workloads in Azure DocumentDB

Native integration with Azure Monitor:

- Metrics from your Azure DocumentDB clusters (CPU, RAM, IOPS, storage, latency) are automatically collected for analysis and dashboards

Diagnostics and resource logs routed to Log Analytics:

- Enables deep query-able visibility into operations

Alerting & proactive notifications:

- Define alert rules on key metrics and hook them into actions

Unified dashboard and cross-resource views:

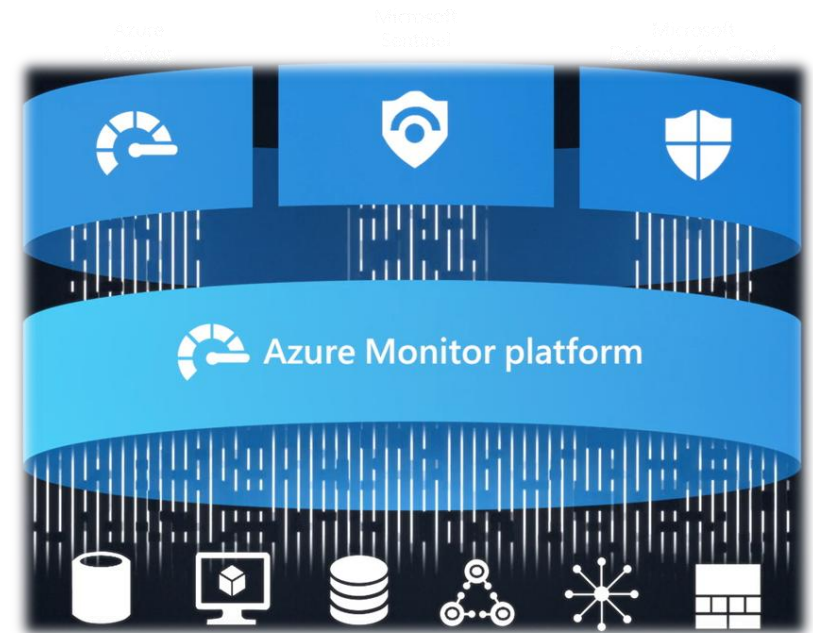
- Monitor your Azure DocumentDB clusters and applications within the same monitoring pane for holistic insights

Detailed troubleshooting:

- Run queries to identify specific error codes

Operational readiness for enterprise workloads:

- Monitoring is not an afterthought, it's built into the architecture



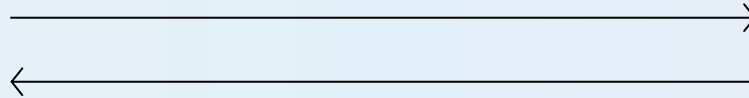
Introducing

LLM-based Index Advisor for DocumentDB

DocumentDB analyzes your workload and query patterns.



DocumentDB analyzes your workload
and query patterns



Personalized index recommendations

Say goodbye to manual troubleshooting
with one-click AI optimization

Introducing Index Advisor for DocumentDB (preview)

Key Features

- One-click AI optimization of your database performance
- Automatically analyzes your workload and query patterns
- Provides meaningful insights and personalized index recommendations with the power of a Large Language Model (LLM)
- Understands the impact of each index, helping you decide whether the index is needed or not

The screenshot displays the Index Advisor for DocumentDB interface. On the left, a sidebar shows the 'DOCUMENTDB CONNECTIONS' list, including 'DOCUMENTDB Local', 'demo@a-bug-bash-vcare.mongocluster.cosmos.azure.com', and 'auser@a-demo-vcare.mongocluster.cosmos.azure.com'. Below this is the 'SERVICE DISCOVERY' section with 'Azure Cosmos DB for MongoDB (RU)', 'Azure Cosmos DB for MongoDB (vCore)', and 'Azure VMs (DocumentDB)'. The main panel shows the 'StoreData/stores' collection with a 'Find Query' button and a 'Query Insights Mock' tab. The 'Query Insights Mock' tab displays execution metrics: Execution Time (2.333 ms), Documents Returned (2), Keys Examined (2), and Docs Examined (10,000). Below these metrics, the 'Optimization Opportunities' section highlights a 'Create Index' recommendation with 'HIGH PRIORITY'. The recommendation states: 'The query performs a COLLSCAN examining 10,000 documents to return only 2, indicating poor selectivity without an index. Recommended Index: { user_id: 1 }. Risks: Additional write and storage overhead for maintaining a new index.' Below this, a 'No Index Changes Recommended' section states: 'The query performs a COLLSCAN examining 50 documents to return 28 (boolean filter selectivity ~56%). A boolean field with over half the collection matching offers low selectivity, so an index on flag alone would not significantly reduce I/O. Execution time is already only 0.02 ms on a 50-document collection, so optimization benefit is negligible.' The bottom section, 'Understanding Your Query Execution Plan', explains: 'Your current query uses a COLLSCAN (collection scan) strategy, which means MongoDB examines all 10,000 documents in the collection to find the 2 matching documents. This is highly inefficient with a selectivity of only 0.02%.' On the right, the 'Query Efficiency Analysis' section shows 'Execution Strategy: COLLSCAN', 'Index Used: None', 'Examined/Returned Ratio: 5,000 : 1', and 'In-Memory Sort: No'. The 'Performance Rating' is 'Poor' with the note 'Only 0.02% of examined documents were returned'. The 'Query Plan Summary' section shows 'Your Cluster' with '100 returned - 100 keys - 100 docs - 120ms' and a 'Show Stage Details' link. The 'Quick Actions' section includes 'Export Optimization Opportunities', 'Export Execution Plan Details', and 'View Raw Explain Output'.

Vector Search in DocumentDB with DiskANN

- **Create vectors** with ML model or embeddings API
- **DiskANN indexing** delivers 10x faster vector search throughput vs. other services¹, helping you accurately retrieve the most relevant neighbors
- Keep all original data and vector data **within the database**
- **Find similar items** based on data characteristics
- Consume the similarity search **results in your AI application**

1. Read the benchmark study:
<https://aka.ms/VectorSearchCosmosvsAtlas>

Common use cases for Vector Search

Vector Search helps you find similar and related items from a vast amount of data



Detecting anomalies



Searching for similar content



Finding related images

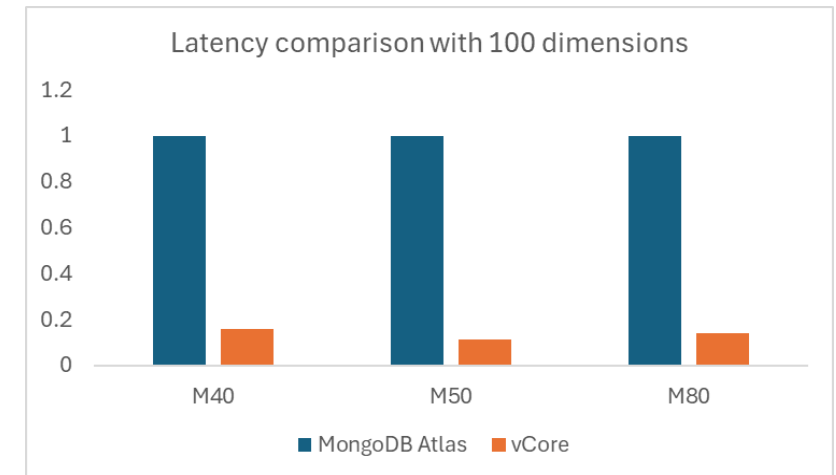
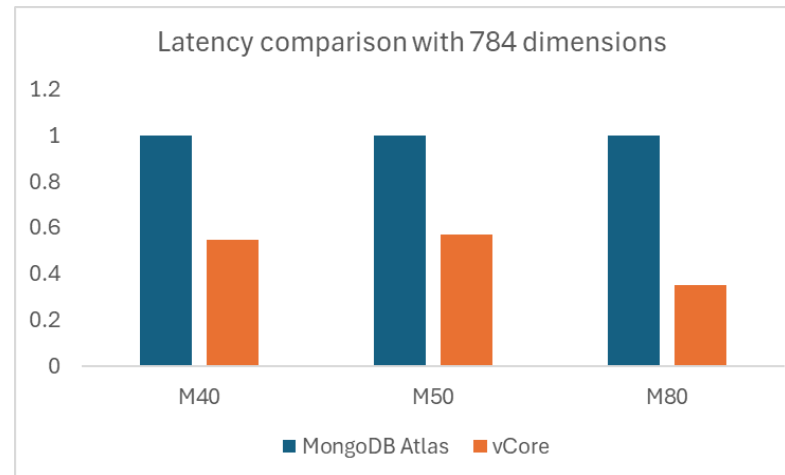
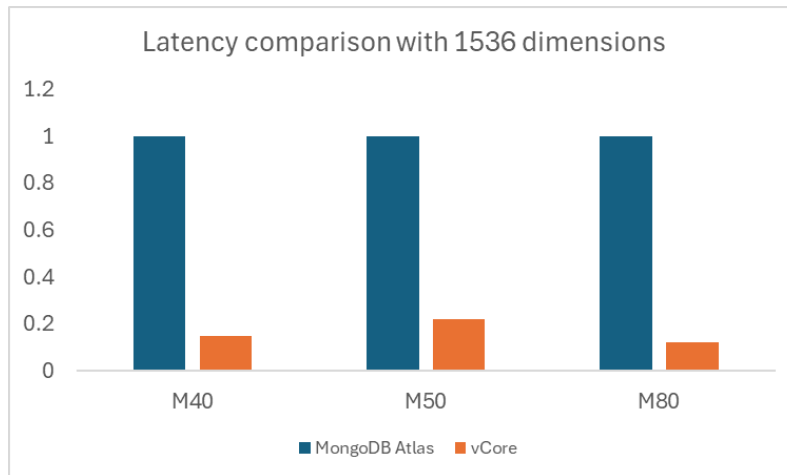
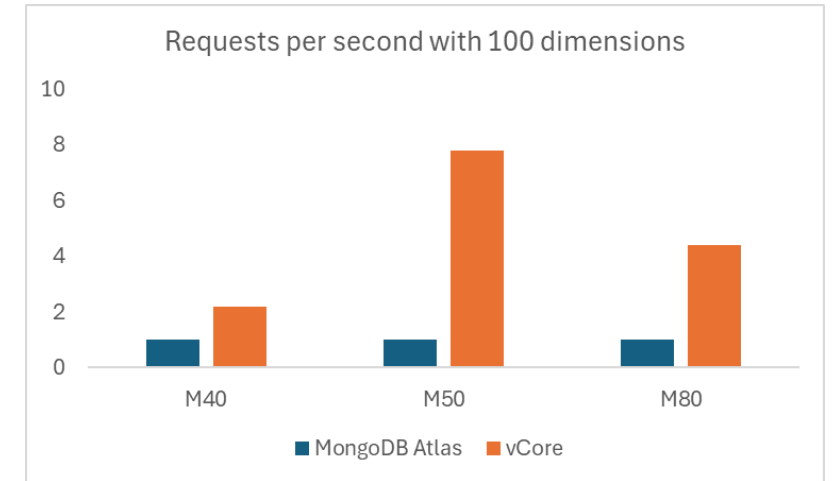
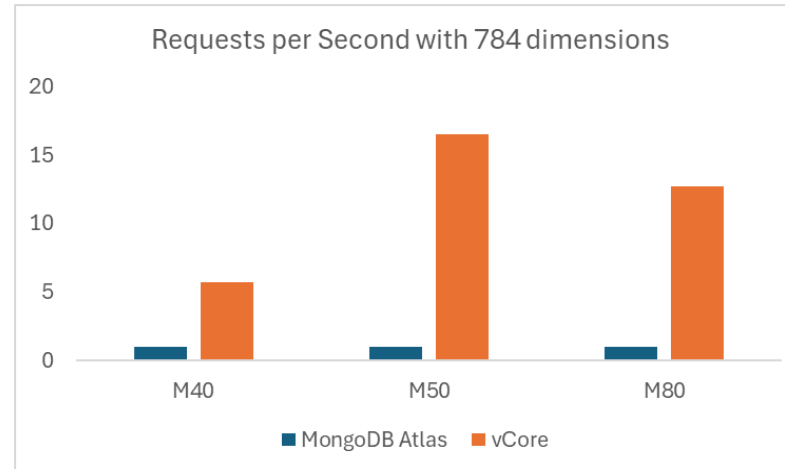
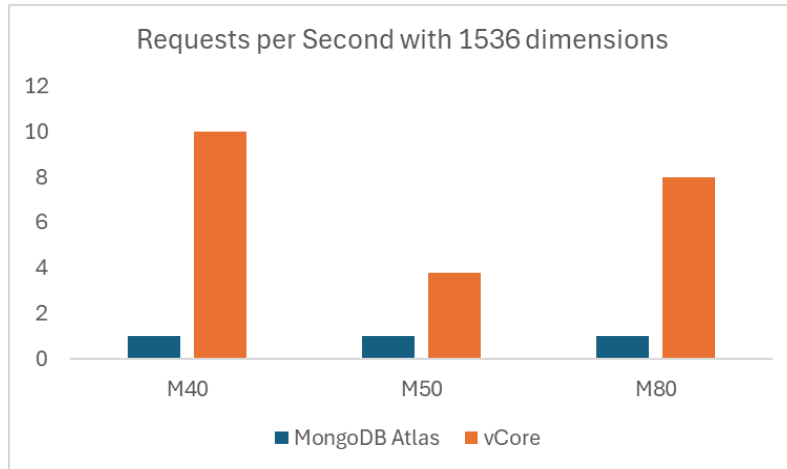


Making personalized recommendations



Answering questions

Azure DocumentDB – Vector Search Benchmarking



Automatic Embedding Generation

Manual Inline Embedding Generation

- Embed via MongoDB commands with AOAI integration
- Supports insert, update, and search operations

Trigger-Based Auto Embedding Generation (Coming Soon)

- Auto-generates embeddings on inserts/updates for a fully integrated workflow

Advanced Full-Text Search in Azure DocumentDB

(gated preview)

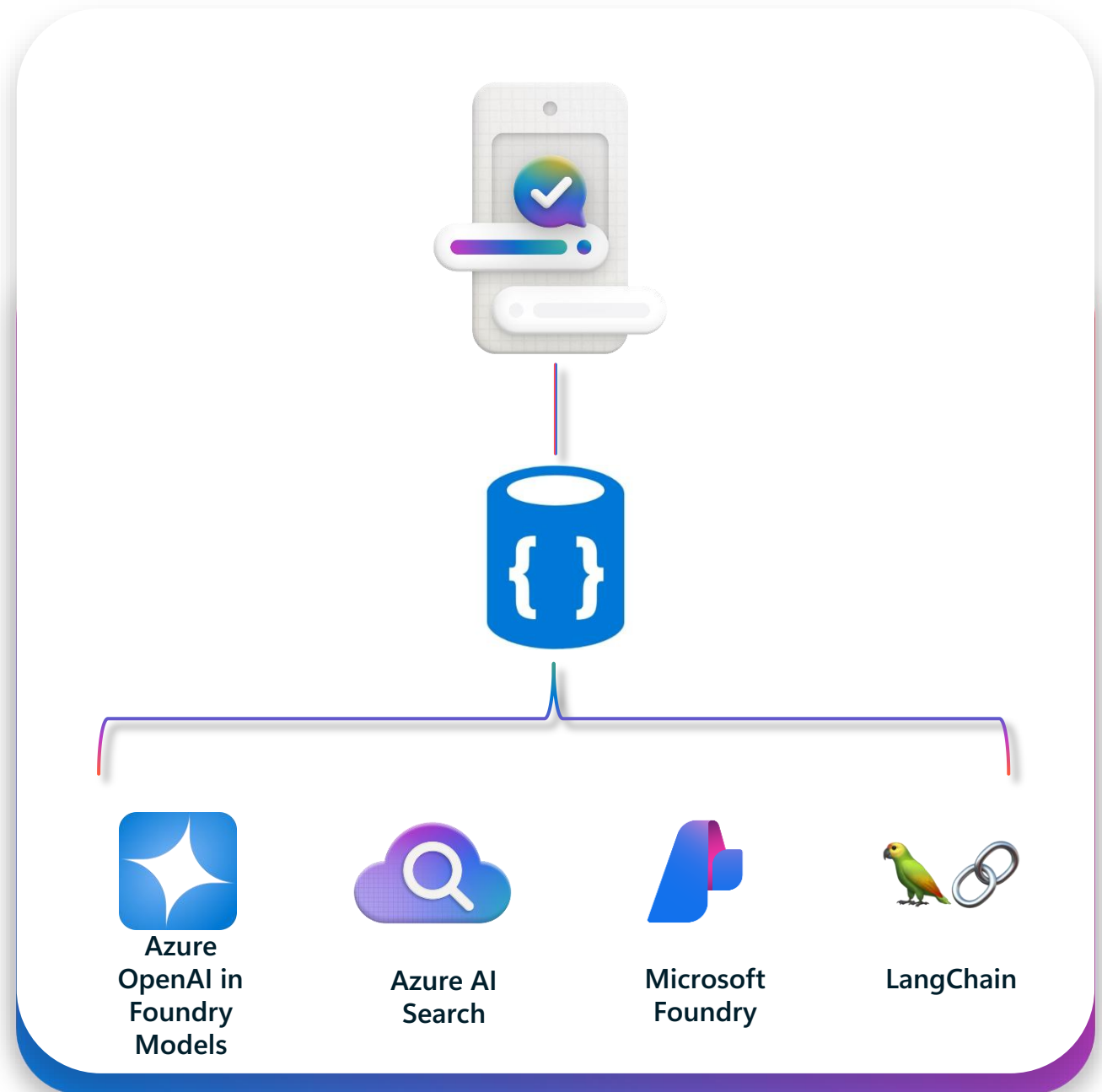
Feature	Description	Impact
 Fuzzy Search	Enables typo-tolerant matching using Levenshtein distance	Improves tolerance for spelling variations and errors
 Proximity Search	Finds words within a configurable distance	Enables phrase-context relevance like "azure" NEAR "document"
 Chinese & Thai Language Support	Adds advanced analyzers for CJK and Thai segmentation	Expands global language coverage for enterprise search
 BM25 Ranking	Implements modern relevance scoring model	Delivers smarter ranking based on term frequency and document length
 \$RankFusion	Combines multiple ranking models (e.g., BM25 + vector)	Bridges keyword search with semantic relevance for AI retrieval

Azure AI integration with Azure DocumentDB

Easily integrate RAG applications with Azure DocumentDB– no coding needed!

Integrations such as a vector store in **LangChain** (Python) to store and retrieve high-dimensional vectors as part of your LLM pipelines.

Use **Microsoft Foundry** to find models like **Azure OpenAI** and build AI solutions faster with prebuilt and customizable models, using your data to innovate at scale.



KPMG sees huge productivity gains and drives new business with generative AI

Azure DocumentDB (formerly vCore-based Azure Cosmos DB for MongoDB) delivers higher quality and speed of query responses

Challenge

- KPMG Australia wanted to develop a private, secure AI assistant with a conversational interface to help employees surface insights from unstructured data

Outcomes

- KymChat is a generative AI assistant that automates mundane tasks and helps employees find data and documents fast.
- KPMG Australia has seen a 50 percent jump in productivity across document drafting, communications, proposal development, and research.
- KPMG clients can also use KymChat with their private organizational data.



The results of our adoption of Azure DocumentDB were startling. Search quality jumped from 50 to 91 percent, and our results were being delivered in under a second.

Robert Finlayson
Principal Product Manager